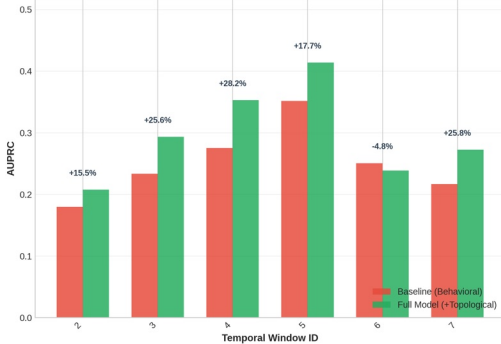
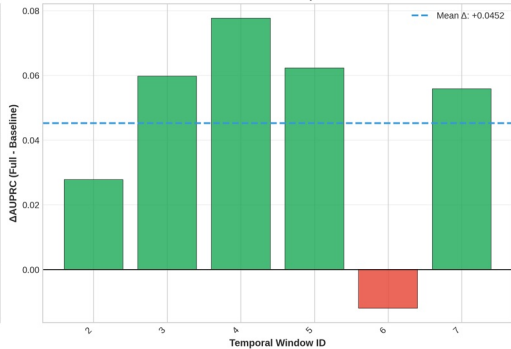


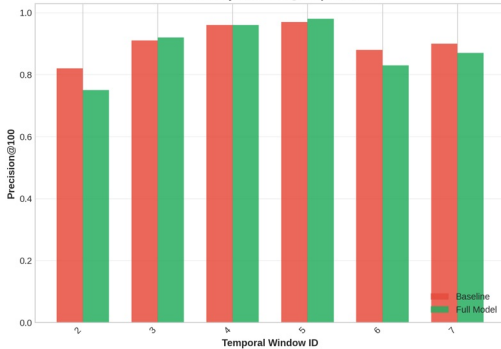
Ablation Study: AUPRC per Temporal Window



Per-Window AUPRC Improvement



Ablation Study: Precision@100 per Window



ABLATION STUDY: STATISTICAL SUMMARY

AUPRC (Area Under Precision-Recall Curve)

Mean Δ : +0.0452 $\pm$ 0.0324
t-statistic: -3.4241
p-value: 1.8755e-02
Significant: ✓ YES ($\alpha=0.05$)

ROC-AUC

Mean Δ : +0.0300 $\pm$ 0.0148
p-value: 4.2431e-03

Precision@100

Mean Δ : -0.0217 $\pm$ 0.0337
p-value: 1.7626e-01

CONCLUSION:

Topological features provide STATISTICALLY SIGNIFICANT improvement in fraud detection performance.